University of Colombo, Sri Lanka

University of Colombo School of Computing

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination - Semester 2 - UCSC AY21[held in March./ Apr. 2025]

SCS1308 — Foundations of Algorithms

(Two (2) Hours)

Answer ALL questions

Number of Pages = 11 Number of Questions = 4

To be completed by the candidate	
Index Number:	

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Important Instructions to Candidates:	To be completed by the examiners
I. Students should answer in English **only** using the space provided in this question paper. II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately. III. Write your index number CLEARLY on every page of this Question paper. IV. Answer ALL questions. This paper consists of Four (4) questions on Eleven (11) pages (including the Cover Page). All questions **do not** carry equal marks. V. Programmable Calculators and any electronic device capable of storing and retrieving text, including electronic dictionaries, smartwatches and mobile phones, are not allowed. VI. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate.	

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Question 1(25 marks)

(a) You are given an array of size n that needs to be sorted by dividing it into two halves and merging them into a sorted data array .

(i). Write the recurrence relation for the above scenario, considering both recursive and non-recursive terms. Express your final answer in terms of T(n).

[5 Marks]

ANSWER IN THIS BOX

(ii) Solve the recurrence equation using recursion tree method and provide the complexity. Note that tree structure including root, depth and how leaves in some level's formation required to obtain full marks.

[10 marks]

ANSWER IN THIS BOX

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(b) Consider the following program that find if an element x exist in an array A[0…n-1].

```
found = False
for i from 0 to n-1:
    if A[i] == x:
        found = True
        break
```

(i). What is the loop invariant of the above code ? [2 marks]

ANSWER IN THIS BOX

(ii) Prove the correctness of the loop invariant for initialization, maintenance and termination phases.

[8 marks]

ANSWER IN THIS BOX

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Question 2 (25 marks)

(a) Define the *"Halting problem"* ? Show that Halting problem is undecidable. Note: You can prove it using an example program.

[10 Marks]

ANSWER IN THIS BOX

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(b) You are given that 3SAT is NP complete. Show that Clique problem is NP hard using reduction. 3SAT can be represented as follows where each clause contains exactly three(3) literals.

$$F = (x_1 \vee \overline{x_2} \vee x_3) \wedge (\overline{x_1} \vee x_2 \vee x_3) \wedge (\overline{x_1} \vee \overline{x_2} \vee \overline{x_3})$$

[15 Marks]

ANSWER IN THIS BOX

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Question 3 (20 marks)

(a) You are given a sorted array A.

A=[1,3,5,7,9,11,13,15,17,19]

(i). Use the jump search algorithm to find the element x=15. Use $\sqrt{n}$ as the jump size. Show all intermediate steps until you find x.

[8 marks]

ANSWER IN THIS BOX

(ii). What are the best and worst time complexities of jump search ?

[2 marks]

ANSWER IN THIS BOX

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(b). Suppose you are designing a system where you need to store **unique student roll numbers** for fast lookup.

Given the following roll numbers:

S={23,45,12,34,56}

Design a **Perfect Hash Function** such that:

a. There are no collisions.

b. Lookups are in constant time

(i). Define your hash function and apply the hash function for each key and build your hash table.

[7 Marks]

ANSWER IN THIS BOX

(ii). According to your selected hash function, what is the lookup time and static insert time complexity ? Explain why prefect hashing would work here ?

[3 Marks]

ANSWER IN THIS BOX

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Question 4 (30 marks)

(a). In depth first traversals, you can visit as deep as possible. Write the pseudocode for pre-order, in-order and post-order transversals. [3 Marks]

ANSWER IN THIS BOX

(b) Draw the final balanced binary search tree after inserting the following sorted sequence of numbers. [1,2,3,4,5,6,7]. Use Day-Stout-Warren Algorithm (DSW) algorithm for tree balancing.

[7 Marks]

ANSWER IN THIS BOX

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(c) Insert the following values [10,20,30,40,50,25] into an initial empty binary search tree and balance using AVL algorithm. To obtain full marks you need to show balancing factors, violation and rotation directions for each step.

[8 Marks]

ANSWER IN THIS BOX

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(d) Create a Red Black (RB) tree by inserting following sequence of numbers: [8,18,5,15,17,25,40,80].

Show detailed steps for your RB tree balancing for the above list; to obtain full marks you need to show node colors, violation and rotation directions in detail.

[12 Marks]

ANSWER IN THIS BOX

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